

DS-265 DLCV Assignment 2: Vision Transformer Analysis on CIFAR-10

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Abstract

This report presents a structured empirical study of Vision Transformers (ViTs) for image classification on CIFAR-10. Seven experiments analyze data efficiency, patch size, token aggregation, positional encoding, attention entropy, overlap in patch extraction, and layer-wise feature quality. Results show that for the chosen tiny ViT and CNN baselines, trained under the same schedule, CNN remains more data-efficient at all data fractions, smaller patch sizes improve ViT accuracy at increased cost, mean pooling is a near-drop-in replacement for the CLS token, learnable positional embeddings slightly outperform sinusoidal ones while removing positional encoding hurts most, overlapping patches yield the strongest accuracy but with large compute and memory overhead, and representation quality improves progressively with depth. These findings expose practical accuracy-efficiency trade-offs and provide guidance for designing ViT variants under compute constraints.

Experimental Setup

Dataset: CIFAR-10 (32×32 RGB images, 10 classes) with standard training augmentation: random crop with 4-pixel padding and random horizontal flip, followed by normalization.

Training setup: batch size 256, learning rate 10^{-4} for ViT and 10^{-3} for CNN, 60 epochs, GPU training.

CNN baseline: ResNet18-style model [1] with 11,173,962 parameters (about 11.17M).

Tiny ViT: image size 32, patch size 4, embedding dimension 192, depth 12, 3 heads, MLP ratio 4, CLS token enabled, learnable positional embeddings [2] unless modified by the experiment, with 5,362,378 parameters (about 5.36M) [3].

Experiment 1: Data Efficiency of ViT vs CNN

Goal: Compare model accuracy under limited training data (5%, 10%, 25%, 50%, 100%).

Table 1: Test accuracy (%) across training data fractions

Data fraction	CNN	ViT	Gap (ViT - CNN)
5%	64.18	39.10	-25.08
10%	74.93	46.12	-28.81
25%	85.58	59.31	-26.27
50%	90.16	68.36	-21.80
100%	92.97	77.16	-15.81

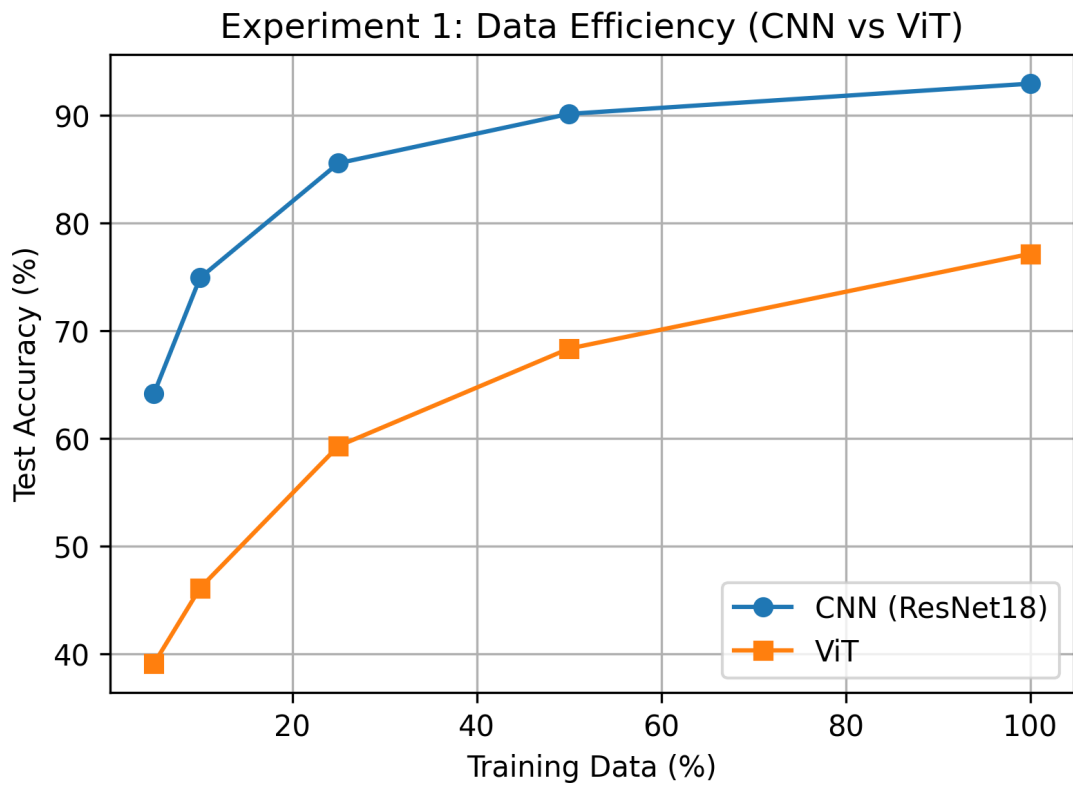


Figure 1: Data efficiency comparison of CNN and ViT

Analysis: Both models improve with additional data, but CNN stays ahead at every fraction: the gap is 25.08 points at 5% and still 15.81 points at 100%. The tiny ViT improves steadily as the data fraction grows, which shows that it is learning useful structure, but under this from-scratch CIFAR-10 setup the convolutional inductive bias remains stronger. Since the CNN and ViT are trained with the same batch size and epoch budget, the comparison is still meaningful from a wall-clock perspective rather than being dominated by a large runtime mismatch. This aligns with observations that small transformers require more carefully tuned designs for competitive performance on small datasets [4].

Experiment 2: Effect of Patch Size

Goal: Study accuracy and training time for patch sizes 4, 8, and 16.

Table 2: Patch size effect on performance and efficiency

Patch size	Test accuracy (%)	Training time (s)
4	77.90	1173.18
8	69.91	896.60
16	58.81	893.95

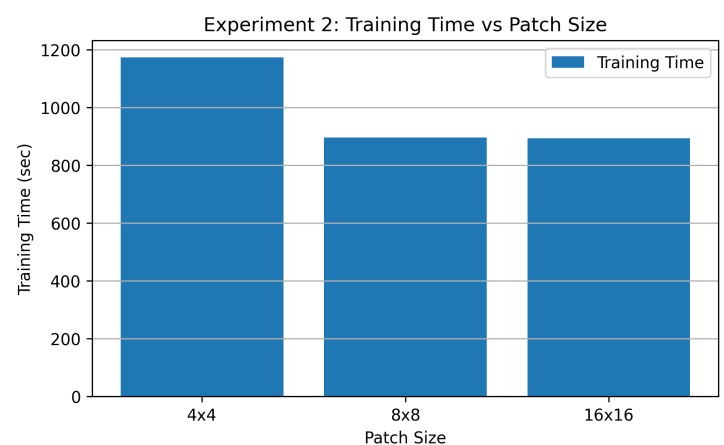
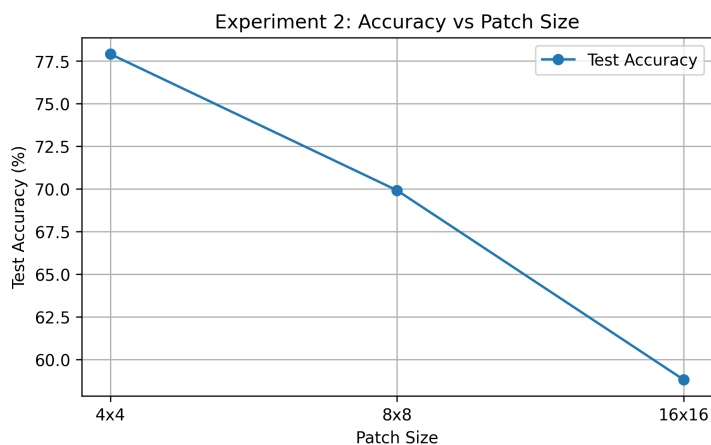


Figure 2: Patch size trade-off: smaller patches improve accuracy at higher cost

Analysis: Patch size 4 gives the best accuracy at 77.90%, improving by 19.09 points over size 16 and by 7.99 points over size 8, but it is also the slowest setting. Smaller patches create more tokens and a finer spatial representation, which helps classification but increases attention cost; the 4-patch model is therefore the best accuracy choice, while 8 and 16 are more economical.

Experiment 3: CLS Token vs Mean Pooling

Goal: Compare standard CLS-token classification with mean pooling over patch tokens.

Table 3: CLS token vs mean pooling

Variant	Test accuracy (%)
ViT with CLS token	77.90
ViT with mean pooling	77.74

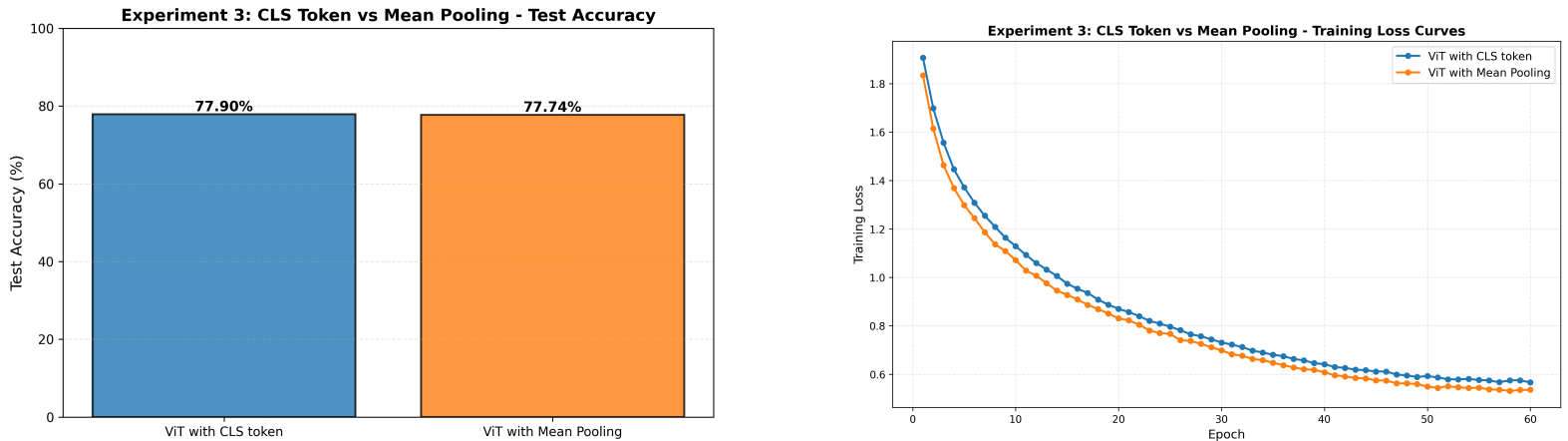


Figure 3: Training loss curves for CLS token and mean pooling

Analysis: Mean pooling is almost indistinguishable from CLS-token pooling: 77.74% versus 77.90%, a difference of only 0.16 points. The loss curves are also very close, which suggests that for this tiny ViT on CIFAR-10, global averaging over patch tokens is a practical replacement for a dedicated CLS token.

Experiment 4: Positional Encoding Ablation

Goal: Compare learnable, sinusoidal, and no positional encoding.

Table 4: Impact of positional encoding strategy

Positional encoding	Test accuracy (%)
Learnable	77.90
Sinusoidal	76.97
None	70.95

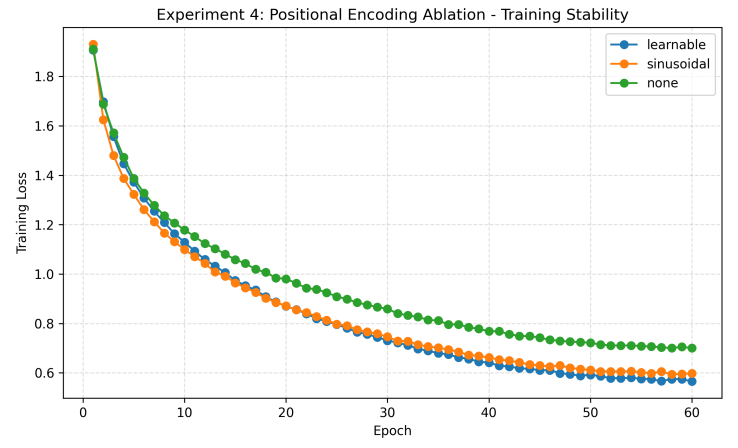
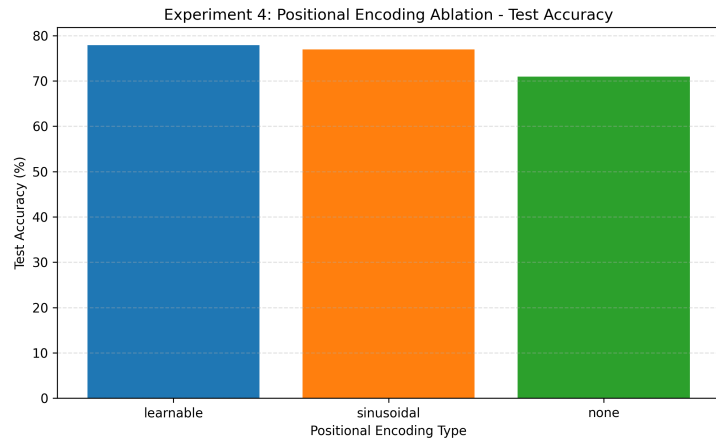


Figure 4: Training stability under different positional encodings

Analysis: Learnable positional embeddings perform best at 77.90%, with sinusoidal embeddings very close behind at 76.97% and the no-position variant dropping to 70.95%. The learnable-vs-sinusoidal gap is only 0.93 points, so the main takeaway is that any explicit positional signal matters, while the exact parameterization matters less. The no-position model also optimizes more slowly, confirming that token order information is essential for ViT performance on CIFAR-10.

Experiment 5: Attention Maps and Entropy Analysis

Goal: Analyze layer-wise and head-wise attention behavior.

Attention entropy is computed as:

$$H = -\sum_i p_i \log p_i \quad (1)$$

where p_i is the normalized attention weight. Larger entropy indicates more diffuse attention, while smaller entropy indicates concentrated attention.

Table 5: Average entropy per layer

Layer	1	2	3	4	5	6	7	8	9	10	11	12
Entropy	3.909	3.594	3.498	3.353	3.501	3.567	3.593	3.486	3.595	3.805	3.734	3.794

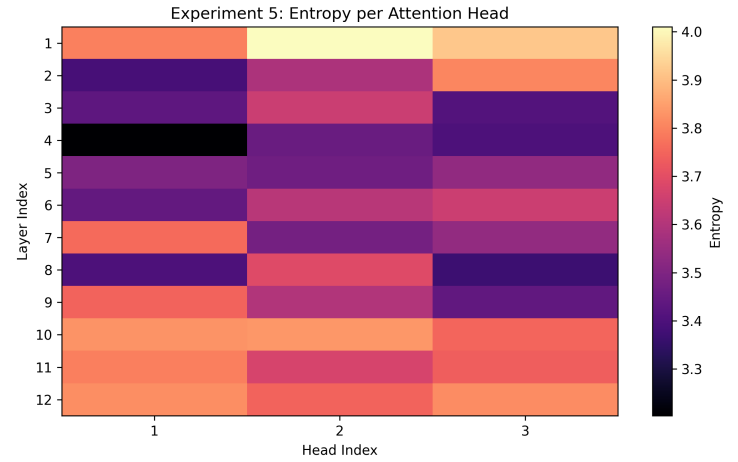
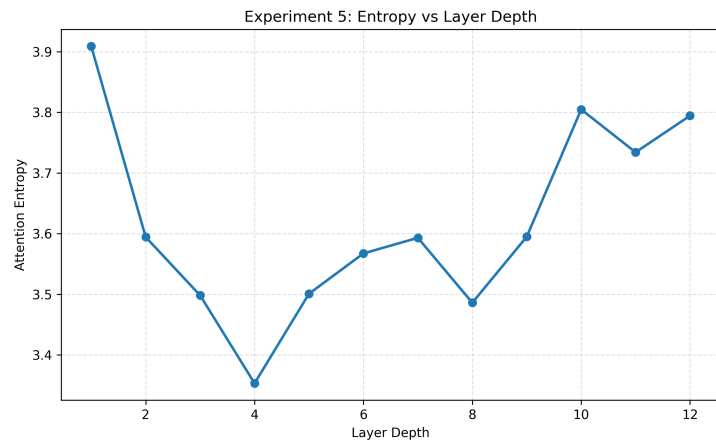


Figure 5: Attention entropy visualizations: layer-wise trend and head-wise heatmap

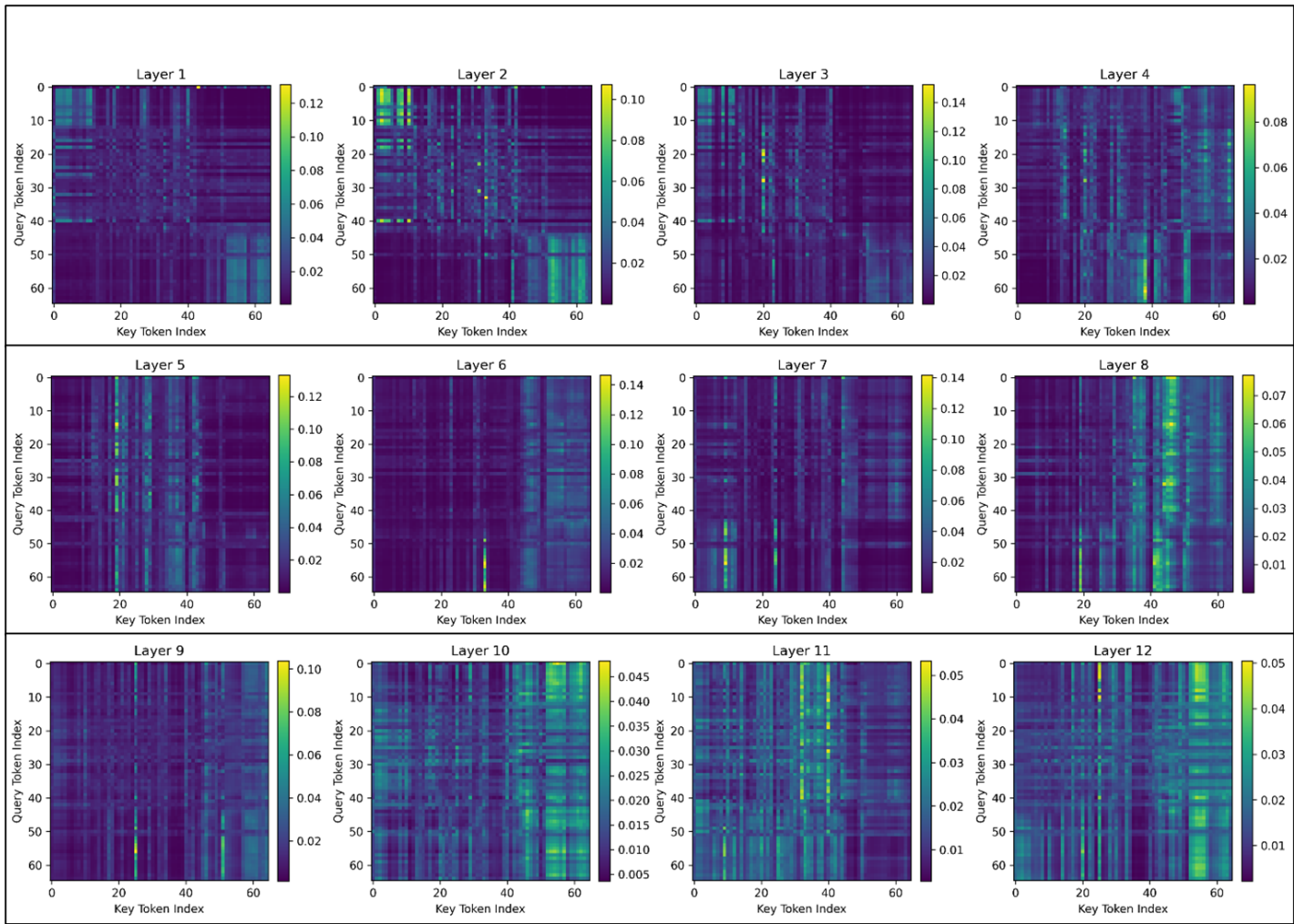


Figure 6: Attention maps across transformer layers

Analysis: Entropy decreases sharply from layer 1 (3.909) to layer 4 (3.353), then fluctuates in mid layers and rises again in late layers (peaking near layer 10 at 3.805 and ending at 3.794 in layer 12). This indicates a progression from broad early attention to focused intermediate routing, followed by broader integration in deeper blocks. Head-wise variations still show clear specialization, so the transformer heads are not behaving redundantly.

Experiment 6: Overlapping vs Non-overlapping Patches

Goal: Evaluate stride-based overlap effects on accuracy and efficiency.

Table 6: Overlap study with patch size 8: stride 8 (non-overlapping) vs stride 4 (overlapping)

Variant	Test accuracy (%)	Train time (s)	Peak GPU memory (MB)
Non-overlapping (stride 8, 16 patches)	69.91	915.28	829.11
Overlapping (stride 4, 49 patches)	78.01	1069.72	2406.11

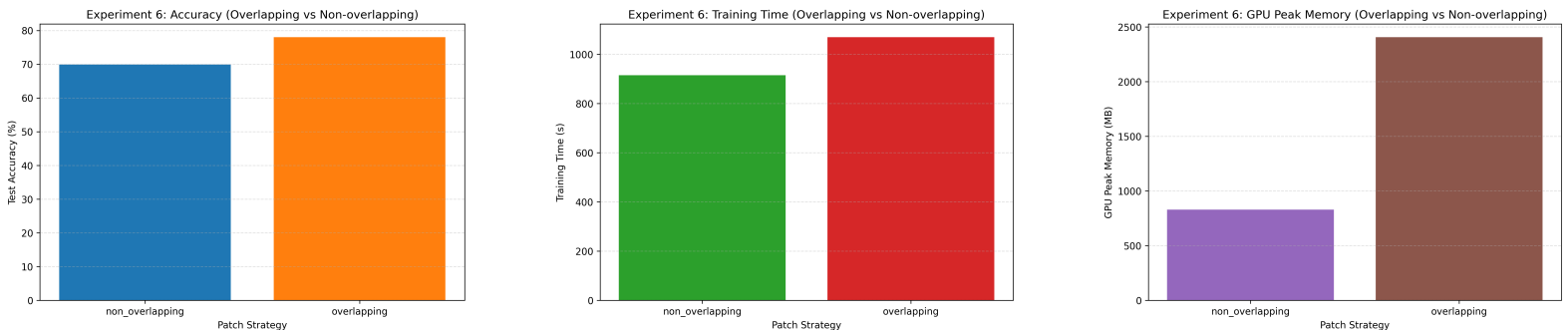


Figure 7: Accuracy-cost trade-off for overlapping patch extraction

Analysis: Overlapping patches improve accuracy by 8.10 points, from 69.91% to 78.01%, increasing token count from 16 to 49 patches ($\approx 3.06\times$). This improvement comes at significant computational cost: $1.17\times$ training time and $2.90\times$ peak GPU memory. The steep memory scaling is the primary trade-off; attention cost grows rapidly with token count. This configuration is practical only when computational budget permits such overhead in exchange for accuracy gains.

Experiment 7: Layer-wise Representation Quality

Goal: Measure linear separability of features from each transformer layer using linear probes.

Table 7: Linear probe accuracy from intermediate ViT layers

Layer	1	2	3	4	5	6	7	8	9	10	11	12
Probe accuracy (%)	37.16	45.43	53.82	61.96	66.72	70.56	73.19	74.94	76.71	77.39	77.94	78.35

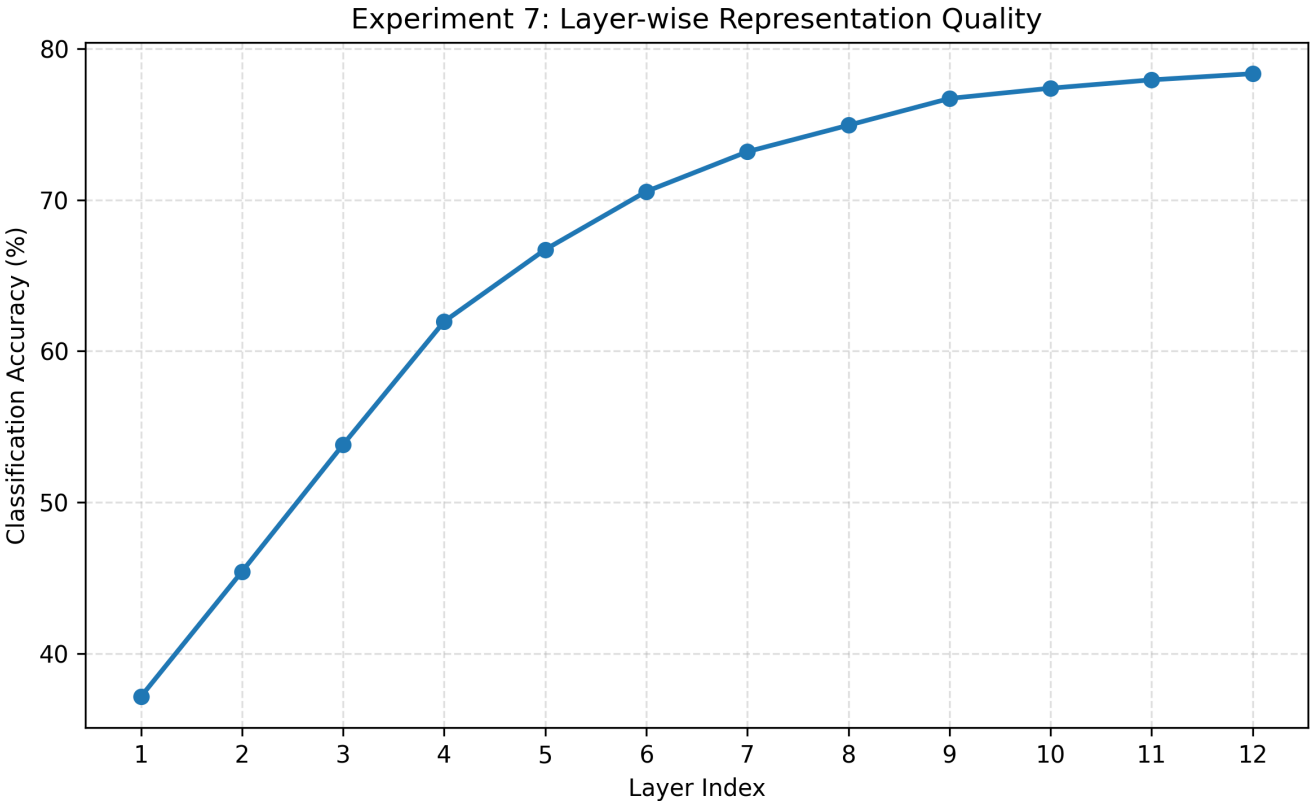


Figure 8: Layer index vs linear probe accuracy

Analysis: Feature quality improves monotonically across all 12 layers, from 37.16% in layer 1 to 78.35% in layer 12. The largest gains appear in early and middle blocks, while later layers (10–12) show smaller incremental improvements, indicating progressive refinement and saturation rather than abrupt representational changes at the end.

Overall Discussion

The seven experiments reveal consistent trends:

- ViT is sensitive to data scale and architectural details (patch size, positional encoding, tokenization strategy).
- Better accuracy often requires significantly higher computational cost (small patches and overlap).
- Several design choices have low-cost alternatives (mean pooling vs CLS) with near-identical performance.
- Internal diagnostics (entropy and linear probes) provide interpretable evidence of how representations evolve across depth.

From a practical perspective, two configurations are particularly meaningful:

1. **Balanced setting:** Non-overlapping patches with sinusoidal positional encoding for improved accuracy without major runtime overhead.
2. **High-accuracy setting:** Overlapping patches when computational budget allows substantial additional time and memory.

Conclusion

This study systematically evaluated ViT behavior on CIFAR-10 through seven controlled experiments. The final results show that while a baseline CNN remains stronger in low-data regimes under the same training budget, ViT performance is strongly improved by finer tokenization and explicit positional encoding. Learnable positional embeddings are slightly better than sinusoidal ones here, and overlap-based tokenization yields the highest accuracy but with heavy resource demand. Layer-wise probing and entropy trends confirm progressively richer and more globally integrated representations in deeper layers. Overall, the experiments provide both performance insights and implementation-level guidance for selecting ViT design choices under different resource constraints.

References

- [1] He, Kaiming, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. “Deep residual learning for image recognition.” *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 2016.
- [2] Vaswani, Ashish, et al. “Attention is all you need.” *Advances in Neural Information Processing Systems* 30 (2017).
- [3] Woo, Sangyun, et al. “TinyViT: Fast pretraining distillation for small vision transformers.” *European Conference on Computer Vision (ECCV)*. 2022.
- [4] Zhang, Ziming, et al. “Powerful design of small vision transformer on CIFAR-10.” *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*. 2024.